Lesson notes · for you

Draw the plans that fit

Optimization · Level 2 · Lesson 2

Time
20 min

Materials
Graph paper (or squares ruled on plain paper); A ruler and a pencil; Coloured pencils for the three lines

Learning goals

- Draw each limit as a line and shade the plans that fit.
- Find the best plan at a corner.
- See how a price change moves the best corner.

Activity: The garden graph

1. Draw (8 min): Draw the axes from 0 to 18 across and 0 to 14 up. Draw land (8 and 8), water (7 across, 14 up) and hours (18 across, 6 up), each in its own colour.
2. Shade (4 min): Shade the area below all three lines. Check a point inside, such as (2, 2), against every limit.
3. Corners (5 min): Mark the five corners and work out 30x + 40y at each. Star the best one.
4. What if (3 min): Work out 50x + 40y at the corners. Which corner wins now, and why did it move?

Let one square be one bed. No graph paper? Rule squares on plain paper with a ruler.

Discussion

- Why is the best plan at a corner and not in the middle?
- Why doesn't extra water help, but extra hours would?
- How would you share the garden work fairly?

Check yourself

- Sketch the shaded shape and star the best corner.

Try it this week
Draw the garden on graph paper: the three lines, the shaded shape and its five corners. Work out the profit 30x + 40y at each corner and star the best. Then try tomatoes at 50 coins: which corner wins now?

Worksheet

Draw the plans that fit

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Name _____ Class _____ Date _____

1 **Your class's garden: tomatoes now sell for 50 coins a bed, and beans still for 40. Does the best corner change?**

◯ No: (3, 5) stays best ◯ Yes: it moves to (6, 2) ◯ Yes: it moves to (0, 0)

2 **The plan 4 tomato beds and 4 bean beds keeps all three limits.**

◯ True ◯ False

3 **Put the steps of finding the best plan on graph paper in order.**

☐ Work out the profit at each corner

☐ Mark the corners of the shaded shape

☐ Pick the corner with the most profit

☐ Draw each limit as a straight line

☐ Shade where every limit is kept

4 **With the usual prices (30 coins a tomato bed, 40 a bean bed), how much does the corner (6, 2) earn?**

◯ 260 coins ◯ 380 coins ◯ 290 coins ◯ 240 coins

5 **Which of these points are corners of the shaded shape? Choose all that apply.**

☐ (7, 0) ☐ (8, 0) ☐ (3, 5) ☐ (4, 4) ☐ (0, 0)

✎ **Where else could you draw a choice with two variables, like the garden? What would the lines be?**

Check your answers

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1 **Yes: it moves to (6, 2)**

Work out 50x + 40y at the corners: (3, 5) gives 350, (6, 2) gives 380 and (7, 0) gives 350.

2 **True**

Land: 4 + 4 = 8, just enough. Water: 8 + 4 = 12, under 14.

3 **Top to bottom: 4, 3, 5, 1, 2**

The lines make the shaded shape, and the best plan is always one of its corners.

4 **260 coins**

30 × 6 + 40 × 2 = 180 + 80 = 260. That is less than (3, 5)'s 290, so with these prices (3, 5) stays the best.

5 **(0, 0); (7, 0); (3, 5)**

(0, 0), (7, 0) and (3, 5) are where two edges meet.